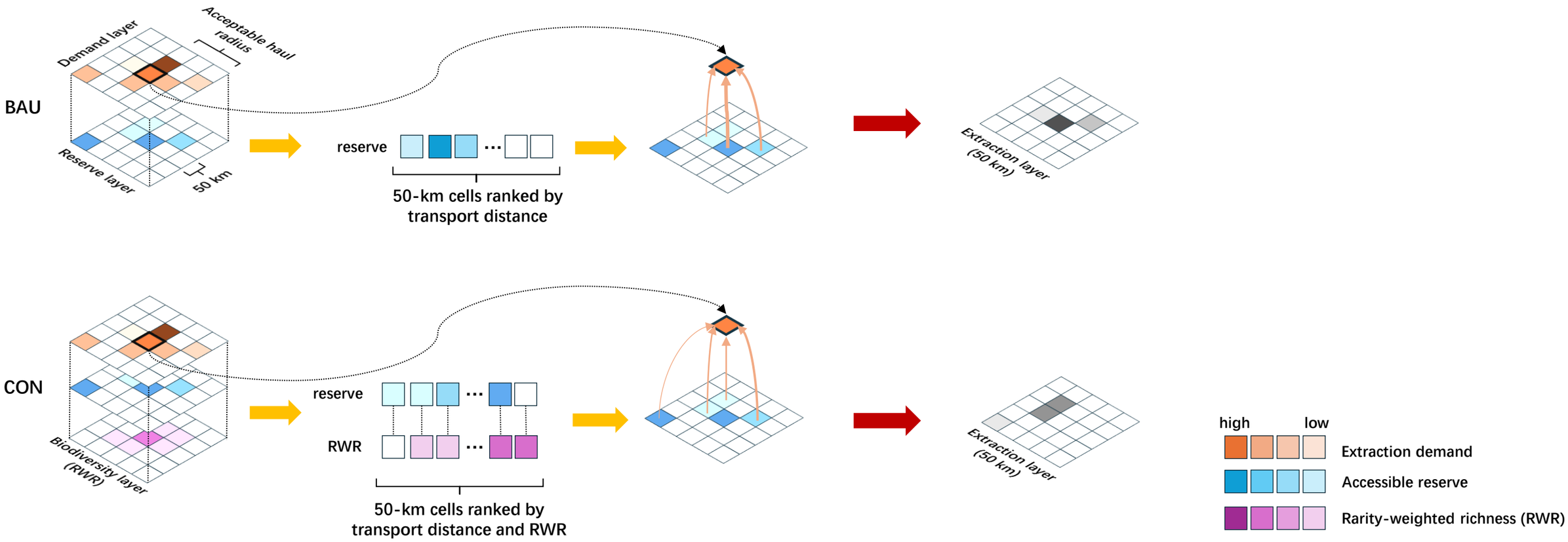
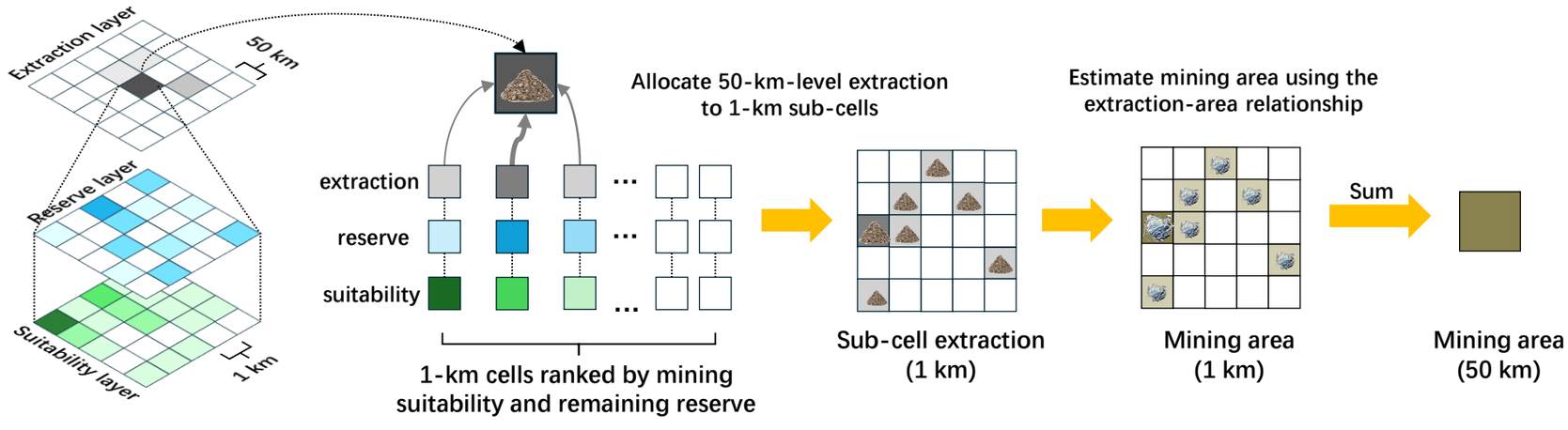
a


BAU
Demand layer
Acceptable haul radius
Reserve layer
50 km
reserve
50-km cells ranked by transport distance
Extraction layer (50 km)
CON
Biodiversity layer (RWR)
reserve
RWR
50-km cells ranked by transport distance and RWR
Extraction layer (50 km)
high
low
Extraction demand
Accessible reserve
Rarity-weighted richness (RWR)
Extraction
Mining suitability
Mining area

b


Extraction layer
50 km
Reserve layer
Suitability layer
1 km
extraction
reserve
suitability
1-km cells ranked by mining suitability and remaining reserve
Allocate 50-km-level extraction to 1-km sub-cells
Sub-cell extraction (1 km)
Estimate mining area using the extraction-area relationship
Mining area (1 km)
Sum
Mining area (50 km)